

Climbing the Ladder: Automating Data Pipelines and Reproducing Baselines in Low-Resource Settings

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Abstract

Sicilian is a low-resource Romance language that the massively multilingual systems now covering it learned from noisy, non-standardised web text, so that what it lacks is no longer coverage but quality; in order to obtain better translations we redevelop the small neural machine translation (NMT) system that Wdowiak (2022) customized for Sicilian. We rebuild the parallel-corpus pipeline on sentence-embedding extraction and alignment (PyMuPDF and LaBSE (Feng et al., 2022)), we assemble a reproducible, deduplicated Sicilian–English dataset with a held-out *literary* test set (not FLORES), and we port the paper’s tokenization and desinence-biasing recipe to a Perl-free stack. We evaluate both from-scratch Sockeye baselines and the pretrained NLLB-200 model (NLLB Team, 2022) adapted with LoRA (Hu et al., 2021): on our harder held-out test, Sicilian→English rises from a 5.27 BLEU floor to 31.43 with a bidirectionally fine-tuned NLLB-1.3B, above Wdowiak’s published baseline (29.1), while the reverse direction nearly doubles (9.89→18.73). We further include Italian as a closely related, higher-resource bridge language (ita_Latn) that NLLB supports natively for it↔scn, and which we evaluate and fine-tune on a frozen WikiMatrix it↔scn split; the full pipeline, dataset assembly and evaluation are reproducible.

1 Introduction

Languages without an army and navy often lack a standard orthography, so that, absent settled spelling rules and conventions, their written texts exhibit great variability, at times even within a single local dialect, which in turn inhibits the ability of conventional metrics to assess model performance (Aeppli et al., 2023). Even a language like Sicilian, for which published dictionaries have presented a consistent orthography for several hundred years (Pasqualino, 1785; Mortillaro, 1876; Nicotra, 1883; Traina, 1868), can exhibit an orthographic diversity that, once incorporated into a training dataset, prevents a model from learning to generate consistent text.

During the 13th century the Sicilian School of Poets at the court of Emperor Frederick II created the first literary standard in Italy, and their poetry inspired Dante to write in his native Florentine, which later became today’s Italian; following in their footsteps, poets such as Antonio Veneziano (1543–1593), Giovanni Meli (1740–1815), Domenico Tempio (1750–1821), Ignazio Buttitta (1899–1997) and Salvatore Camilleri (1921–2021) continued the literary tradition, writing in a fairly consistent manner. Writing, however, is not speaking: Giuseppe Pitre (1841–1916) developed a parallel popular tradition that recorded the common people in their own words and in the sound of their own words.

One can hear echoes of the popular tradition in the daily speech of millions of people in Sicily, Calabria and Puglia, as well as in diaspora communities abroad, and echoes of the literary tradition in books, journals and pages of the internet; the publicly available Sicilian text, however, exhibits an orthographic diversity that once prevented practitioners from assembling datasets of parallel text and, for a time, kept Sicilian out of every major translation service. Today Sicilian is included in massively multilingual models such as NLLB-200 (NLLB Team, 2022), but the coverage is of low quality, since

the Sicilian training text was mined from non-standardised web sources and the model has, in effect, learned a non-standard orthography (Section 4); the barrier for Sicilian has thus shifted from absence to quality.

High-quality material does exist, above all in the bilingual literary journal *Arba Sicula* (published since 1979), which Wdowiak (2022) used to build the first Sicilian neural machine translation (NMT) system; that work showed that a fairly good translation model can be trained in a low-resource setting, at the cost of a labour-intensive process in which the sentences were aligned by hand with a recipe tailored to the low-resource, morphologically rich case. The present work rebuilds that effort with maintained, mostly CPU-only tooling, and seeks better quality through modern pretrained models.

Contributions.

- An automated extraction and alignment pipeline for the *Arba Sicula* PDFs (PyMuPDF and LaBSE (Feng et al., 2022)), replacing the legacy pdflatex+hunalign flow, with thresholds tuned against Wdowiak’s hand-aligned gold.
- A reproducible, deduplicated, provenance-tracked dataset with a held-out literary test set (not FLORES), and a BLEU/chrF evaluation harness.
- The paper’s “lever B” recipe (Napizia tokenizer + desinence-biased subwords) ported to a Perl-free pipeline, and a linguistic-feature lever (lemma source factors).
- NLLB-200 zero-shot and LoRA fine-tuning that beats the published baseline on our harder test set, with a bidirectional adapter that also rescues the reverse direction.

2 Background

NMT models translation as a conditional language model, in which an encoder reads the source $x = (x_1, \dots, x_n)$ and a decoder factorises the target from left to right, $p_\theta(y | x) = \prod_t p_\theta(y_t | y_{<t}, x)$, the model being trained by maximising the log-likelihood of a parallel corpus and decoded with beam search; the dominant parameterisation is the Transformer (Vaswani et al., 2017), which replaces recurrence with scaled dot-product attention, $\text{Attention}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})V$, run in parallel heads.

With little data the default Transformer overfits, so the low-resource recipe of Sennrich and Zhang (2019) prescribes a smaller, heavily regularised model, with high dropout (Srivastava et al., 2014) and small subword vocabularies; subword segmentation by byte-pair encoding (Sennrich et al., 2016) gives an open vocabulary and shares statistics across word forms, which is essential for a morphologically rich language. Further gains come from linguistic input features (Sennrich and Haddow, 2016), from back-translation of monolingual data (Sennrich et al., 2015), and from multilingual training with a target-language token (Johnson et al., 2016) that enables bridging through related high-resource languages (Fan et al., 2020; Arivazhagan et al., 2019); low-resource MT remains hard for exactly these reasons (Koehn and Knowles, 2017).

3 Data Pipeline

3.1 Extraction from PDFs

We extract clean text from the *Arba Sicula* issue PDFs with PyMuPDF (no pdflatex round-trip). Each page is assigned a language from its share of Sicilian stop-words and diacritics; the journal sets Sicilian and English on facing pages, giving a natural even/odd prior, so we form facing candidate pairs. Pages are segmented into sentences with pysbd (per language), filtering “furniture” (all-caps headers, page numbers, captions). The procedure recovers parallel pages the manual selection had skipped.

Table 1: Unified dataset.

split	pairs	source
train	27,386	Arba Sicula + NLLB
valid	1,000	Arba Sicula
test	1,000	Arba Sicula (literary)
it-scn (aux)	11,389	WikiMatrix

3.2 Sentence alignment

We embed every sentence with LaBSE (Feng et al., 2022), a language-agnostic dual encoder, and build the cosine-similarity matrix $S_{ij} = \langle u_i, v_j \rangle / (\|u_i\| \|v_j\|)$. A facing pair is confirmed only if its mean best-match similarity exceeds τ_{page} ; sentences are then aligned by a monotonic dynamic program over S allowing 1–1, 1–2 and 2–1 moves with a null penalty, in the spirit of Vecalign (Thompson and Koehn, 2019) but without a bilingual dictionary or hunalign. The thresholds $(\tau_{\text{page}}, \tau_{\text{sent}}) = (0.50, 0.40)$ were tuned against Wdowiak’s hand-aligned gold (issues AS41–42); the pipeline recovers 14,245 sentence pairs from 44 issues, comparable to his manual alignment, and the *unaligned* English is retained as an in-domain monolingual pool for back-translation.

3.3 Additional sources

We add the author’s curated “Good-Sicilian-in-NLLB” set (LASER3-mined sentences (Heffernan et al., 2022; NLLB Team, 2022) rescored and filtered for quality and for Corsican contamination), a small curated “Good-Sicilian-from-WikiMatrix” set, and the WikiMatrix it-scn subset (Schwenk et al., 2021) as an auxiliary Italian bridge.

3.4 Unified dataset

Sources are cleaned, deduplicated across one another, and split deterministically. Then a 1,000/1,000 valid/test split is *frozen* once, held out from Arba Sicula (which provides the high-quality literary standard that FLORES does not), so that every later result stays comparable (Table 1).

4 Standardization

Although Sicilian does not have a single official orthography, a common literary standard has existed for several hundred years and provides a rough consistency, which we can refine because the majority of the Sicilian sentences in the NLLB dataset were written in that standard. Roughly one-third of the sentences the NLLB team collected were nonetheless written in local dialect, some of them in Corsican, and very few of the pairs carried a valid translation; we suspect that these problems trace back to seed data that is itself inconsistent with the literary standard.

4.1 Why Sicilian lives in the Romance neighbourhood

A pretrained multilingual encoder needs no explicit Sicilian support to *read* Sicilian. The representations it learns are not a shared interlingua that maps every language into one abstract, universal meaning space; instead they cluster languages by linguistic similarity (Kudugunta et al., 2019) and organise around surface-form cues (Verma et al., 2026). For Sicilian and Italian, which are close on both counts, the two accounts agree: Sicilian lands in the Italian–Romance region either way.

The surface form of literary Sicilian is close to Italian, so reading it works well. On the multi-parallel FLORES-200 devtest (NLLB Team, 2022), LaBSE (Feng et al., 2022) retrieves the correct English translation of a Sicilian sentence 99% of the time (precision@1, Sicilian→English), one point behind *supported* Italian (100%); and, discounting English (LaBSE’s pivot, to which everything is close), each Sicilian sentence’s nearest neighbour is overwhelmingly Italian (18%, twice the next Romance

language). In lexicon and spelling Sicilian is close enough to Italian that the model treats it almost as a variety of Italian.

The seed-data failure, however, is in the *opposite* direction, namely the retrieval of the Sicilian translation of an English sentence. Romance languages share a lot of vocabulary and grammar, so when a model retrieves a Sicilian, Italian or Corsican translation of an English sentence, the translation that it selects will reflect the surface-form cues in its training data. Sometimes the NLLB seed data’s surface form reflects the literary language in Sicilian dictionaries, but other times it reflects local pronunciations, which (we suppose) is what caused their model to frequently retrieve Corsican-language sentences when assembling the NLLB’s Sicilian corpus.

Since multilingual models encode script and surface form over deeper linguistic structure (Verma et al., 2026), the two directions diverge for the same reason: an Italianate surface form retrieves the right English, while Corsican-looking seed data makes the model generate the wrong Sicilian. The proximity is thus double-edged: it is exactly what lets our own LaBSE-based aligner (Section 3.2) operate on an officially unsupported language, but it offers no protection against near-language contamination when a corpus is assembled by transfer alone.

Alignability, in turn, tracks *standardness*: scoring the LaBSE cosine of Wdowiak’s hand-aligned gold pairs per text, standard literary prose (Cipolla’s *Don Chisciotti*) aligns at cosine 0.76, with 91% of its pairs clearing our production threshold, whereas dialect poetry aligns at only 0.64 (80% recovered). The non-standard material is thus at once the larger source of noise in the collected data and the harder to recover, so that curating toward the literary standard improves quality on both counts; rolling short verses into longer lines recovers much of the poetry gap (mean cosine 0.65 $\rightarrow$ 0.76, recovery 84% $\rightarrow$ 99% when merging up to three verses), a cheap corrective for the poetic register.

A translation model needs sentences written in the Sicilian literary language together with their translations, so we back-translated all of the Sicilian sentences into English and Italian and then scored each resulting pair on the task of translation into Sicilian; since a sentence written in standard Sicilian scores better than one written in local dialect, this forward scoring of the back-translations lets us identify the sentences written in the standard. The procedure yielded 549,486 back-translations that simulate English-to-Sicilian translation and 689,840 that simulate Italian-to-Sicilian translation.

5 Tokenization and Subword Segmentation

Because the majority of the Sicilian sentences in the NLLB dataset were written in a common literary standard, we can refine them for consistency. We first normalise the text toward the standard established by Cipolla and augmented by Wdowiak for MT clarity, i.e. strict L on the articles (*lu/la/li*), h on *aviri* (*haiu, havi, hannu*), CI for *çi* (*çiuri* $\rightarrow$ *ciuri*) and R retained before a clitic (*farlu*); our `normalize_scn.py` applies the documented spelling rules (e.g. *cchiù* $\rightarrow$ *chiù*, *io/iu/eu* $\rightarrow$ *jo, non* $\rightarrow$ *nun*) at a “std” level, and additionally uncontracts and ASCII-folds at a “full” level. Since NLLB’s own tokenizer was trained on un-normalised web Sicilian, whether this normalisation helps the pretrained model is an empirical question, and we ablate it (*scn* $\rightarrow$ *en*, identical English references): raw 30.60, std 30.98, full 30.98 BLEU (chrF 56.31 / 56.63 / 56.50). The light “std” standardisation gives a small consistent gain (and the best chrF), whereas the aggressive “full” preprocessing does not improve over it, which is consistent with a pretrained tokenizer that expects un-normalised input; orthographic normalisation is thus only a minor lever for NLLB ($\sim +0.4$ BLEU), so we adopt std by default and conclude that the larger residual relative to the state of the art is data curation, not spelling.

We re-implement the Napizia tokenizer in pure Python (verified identical to the original Perl): it ASCII-folds accents and uncontracts Sicilian (e.g. *ô* $\rightarrow$ *a lu*, *dû* $\rightarrow$ *di lu*), which cuts sparsity. “Lever B” then biases the joint *scn*-*en* BPE (Sennrich et al., 2016) toward the textbook desinenes catalogued in the resources Wdowiak assembled to standardise Sicilian (Dieli’s *Sicilian Vocabulary*, Cipolla’s *Mparamu lu sicilianu* (Cipolla, 2013), Bonner’s grammar (Bonner, 2001) and the *Chiù dê Palora* dictionary), so that morphological endings survive as units; for instance Wdowiak’s *Carinisi are dogs!* segments to *car@@ in@@ isi are dogs !*. This cut the vocabulary from 14,000 to 2,000 units and produced his largest BLEU gains, and BPE-dropout (Provilkov et al., 2020) adds subword regularisation by sampling segmentations during training.

Table 2: Our models, our test set (scn→en).

model	BLEU	chrF
floor (copy source)	5.27	25.40
Sockeye baseline	5.54	28.28
+ lever B (tok. + desinences)	7.24	29.52
+ more data	9.79	33.82
+ lever D (lemma source factors)	10.85	35.22
NLLB-200 1.3B, zero-shot (raw)	29.02	55.23
NLLB-200 1.3B, LoRA fine-tuned, scn→en only	31.16	56.79
NLLB-200 1.3B, LoRA <i>bidirectional</i>	31.43	56.94
+ back-translation (7.5k synthetic)	31.60	57.21

6 Models

The from-scratch baseline is a Sockeye-3 (Hieber et al., 2017) small Transformer (3 layers, model size 256, 4 heads, feed-forward 1024, weight tying, dropout 0.4, joint BPE $\approx 4k$; $\sim 6.6M$ parameters), trained on CPU. On top of it we stack lever B (tokenization + desinences), more parallel text, and “lever D” linguistic input features (Sennrich and Haddow, 2016): a lemma source factor summed into the source embedding, $e_j = E_{\text{word}}(w_j) + E_{\text{lemma}}(\ell_j)$, whose small (~ 900) vocabulary generalises across inflected forms.

The pretrained model is NLLB-200 (NLLB Team, 2022), a massively multilingual encoder–decoder that supports Sicilian (scn_Latn) and is steered by forcing the first decoder token. We adapt the 1.3B variant with LoRA (Hu et al., 2021): freezing W_0 and learning a low-rank update $W = W_0 + \frac{\alpha}{r}BA$ ($r=32$, $\alpha=64$) on the attention projections, with the base in fp16 and the adapters in fp32. Training *both* directions at once (scn→en and en→scn) yields a single adapter that serves both and rescues the weak reverse direction.

7 Evaluation

We score with sacreBLEU (Post, 2018), i.e. BLEU (Papineni et al., 2002) (tok: 13a) and chrF (Popović, 2015), on the frozen literary test set rather than on the NLLB/FLORES set, since the FLORES Sicilian uses the post-2017 orthography and a different, non-literary domain and is thus not representative of the Arba Sicula material; the Sockeye rows are scored in tokenized space (the same space as the upstream system, the raw-text floor being 5.27 BLEU), while the NLLB rows are scored on raw text.

8 Results

On the frozen held-out test set (1,000 literary pairs), the Sicilian→English results are reported in Table 2: the Sockeye levers stack from 5.54 to 10.85 BLEU on CPU alone, whereas the pretrained NLLB-1.3B wins decisively even zero-shot (29.02), and LoRA fine-tuning lifts it to 31.43, above Wdowiak’s published baseline (29.1) on our harder test.

In the reverse direction (en→scn), the same bidirectional adapter lifts NLLB-1.3B from 9.89 (zero-shot) to 18.73 BLEU / 49.96 chrF, nearly doubling the weak direction at no cost to scn→en and giving a usable two-way model. This is still below the published en→scn baseline (25.1); back-translation is the next lever.

Italian directions (it↔scn). Italian is a closely related, much higher-resource Romance language, and NLLB supports it↔scn natively; we hold out a frozen 1,000-pair it↔scn test split from the Wiki-Matrix data, evaluate NLLB-1.3B zero-shot, and then fine-tune all four directions jointly (scn↔en and it↔scn) into a single multilingual adapter, a bridge in the sense of Johnson et al. (2016) and Fan et al.

Table 3: Trilingual model: all directions, zero-shot vs. the multilingual fine-tune (BLEU / chrF, frozen test sets).

direction	zero-shot	multilingual FT
scn→en	29.02 / 55.23	30.75 / 56.57
en→scn	9.89 / 41.12	17.49 / 48.78
it→scn	17.61 / 44.77	26.97 / 51.69
scn→it	42.20 / 59.04	43.47 / 59.79

(2020) (Table 3). Italian is by far the easiest pair, as one would expect for a close, high-resource neighbour on in-domain WikiMatrix text: scn→it reaches 43.5 BLEU and it→scn climbs from 17.6 to 27.0. The multilingual adapter costs a little on scn↔en (30.8/17.5 against the dedicated 31.4/18.7), so that one can ship either the trilingual all-rounder or the specialised pair.

Relation to the state of the art. We do not beat the state of the art. On his own in-domain, hand-selected test set Wdowiak’s reverse-training system reports En→Sc/Sc→En BLEU of 45.1/48.6 and It→Sc/Sc→It of 61.4/62.9, well ahead of our 18.7/31.6 and 27.0/43.5 on every direction (his earlier marks were 25.1/29.1 for the baseline and 35.0/36.8 augmented). These numbers are not a head-to-head, since the test sets differ, but the gap is too large to be explained by that alone: on Sc→en we only clear his baseline (31.6 against 29.1), on a harder test, and we trail elsewhere. His reverse-training, we remark, builds a custom pretrained model from tens of millions of pairs (forward-translation, then back-translation, in three stages) before fine-tuning on *Arba Sicula*, a pipeline infeasible to reproduce on a single GPU; NLLB-200, however, already provides that massive multilingual pretraining, so that our fine-tune is the analogue of his stage 3. We applied the feasible parts of his recipe on top of NLLB and each adds little, i.e. the multilingual Italian bridge, orthographic normalisation (+0.4 BLEU) and back-translation (+0.17, capped by our 7.6k in-domain monolingual pool against his ~550k); the method gap is therefore largely closed, and the residual is data, namely his hand-curated, standardised corpus, far more back-translation, and an in-domain test set, which argues for combining his corpus with our pipeline rather than competing with it.

9 Conclusion and Future Work

We have presented a reproducible, Perl-free path from *Arba Sicula* PDFs to a Sicilian–English NMT system, comprising an automated extraction-and-alignment pipeline, the paper’s tokenization and desinence recipe with an added linguistic-feature lever, and NLLB-200 adapted with LoRA. On our harder literary test set the best model reaches 31.6 BLEU (scn→en), above the published baseline but well below Wdowiak’s reverse-training state of the art; having applied the feasible parts of his recipe on top of NLLB (multilingual Italian bridging, orthographic normalisation and back-translation), each for only a small gain, we conclude that the residual is data, namely his hand-curated corpus, far more back-translation, and an in-domain test set, which makes a concrete case for combining his corpus with our pipeline. We also release the serving code (a local web app and a Telegram bot, a FastAPI service over the NLLB adapter); all code, the data pipeline and the evaluation harness are released for reproducibility.

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